

# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\mathcal{I}_D^*(\lambda) = \sup_{\gamma \in \mathcal{P}_D} I_{D,\lambda}[\gamma] = \sup_{\pi \in \Gamma(\Pi, \varphi_\lambda)} \int_{\mathcal{V} \times \mathcal{V}} \varphi_\lambda d\pi = \sup_{\pi \in \Pi(\mu_1, \dots, \mu_L)} \int_{\mathcal{V}} g_\lambda d\pi.$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$G_{\mu\nu}^{(p)} = \eta_{\mu\nu} + \partial_\mu \phi^i \partial_\nu \phi^i - \bar{\lambda}(\Gamma_\mu + \Gamma_i \partial_\mu \phi^i) \partial_\nu \lambda - \bar{\lambda}(\Gamma_\nu + \Gamma_i \partial_\nu \phi^i) \partial_\mu \lambda + \bar{\lambda} \Gamma^m \partial_\mu \lambda \bar{\lambda} \Gamma_m \partial_\nu \lambda,$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$\left(\frac{\det'(\Delta_v)}{\det(\Delta_v^{(0)})}\right)^{-\frac{1}{2}}\left(\frac{\det(\Delta_A)}{\det(\Delta_A^{(0)})}\right)^{-1}\left(\frac{\det(\Delta_c)}{\det(\Delta_c^{(0)})}\right)^{+1}=\left(\frac{\det'(\Delta_-)}{\det(\Delta^{(0)})}\right)^{-1}$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\langle \alpha_1, \alpha_2, ..., \alpha_{2N}, t | U_L | \alpha'_1, \alpha'_2, ..., \alpha'_{2N}, t \rangle = R_{\alpha_1 \alpha_2}^{\alpha'_2 \alpha'_3} R_{\alpha_3 \alpha_4}^{\alpha'_4 \alpha'_5} ... R_{\alpha_{2N-1} \alpha_{2N}}^{\alpha'_{2N} \alpha'_1}$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$P^- = \frac{1}{2} \int dx^- d^2 x^\perp : \left[ -\phi \partial^{\perp 2} \phi + \sum_{i=1}^2 \left( \bar{\psi}_m^{(i)} \frac{-\partial^{\perp 2} + m^2}{i \partial^+} \gamma^+ \psi_m^{(i)} + 2g \phi \bar{\psi}_m^{(i)} \psi_m^{(i)} + g^2 \bar{\psi}_m^{(i)} \phi \frac{\gamma^+}{i \partial^+} \phi \psi_m^{(i)} \right) \right] : .$$